

Instructions for Exercise 6

1. This exercise is graded passed/failed. The early-bird deadline is **16.4.2025**. The final deadline is **16.4.2025**.

To pass the course, the student must complete 6 weekly exercises. However, if the student returns each exercise within the early bird date, 5 exercises are sufficient to pass the course.

2. Return 2 pdf-files
 - i. A pdf-file with the written exercise (1 page) and answers to the questions in the Python files
 - ii. A pdf-file with the Python exercises

A. Written exercise (Individual work)

Write a total of one page (400-500 words) on the following topics. Don't write about things that are already trivial to you. The aim of this exercise is to learn new things.

1. Ethical issues: In Python Exercise 6.1, we will estimate the price of housing based on 12 features. These features are: per capita crime rate by town; proportion of residential land zoned for lots over 25,000 sq.ft.; proportion of non-retail business acres per town; Charles River dummy variable; nitric oxides concentration; average number of rooms per dwelling; proportion of owner-occupied units built prior to 1940; weighted distances to five Boston employment centres; index of accessibility to radial highways; full-value property-tax rate per 10,000 dollars; pupil-teacher ratio by town; lower status of the population; median value of owner-occupied homes in 1000's dollars.

What are the ethical issues involved in using machine learning with such data? What kind of variables would you use in estimating the price of housing? What kind of other ethical issues are there in using ML in finance? (algo-trading, credit-scoring etc.)

2. Quantum Machine Learning in Finance: Investigate the potential impact of quantum computing on financial modeling and optimization tasks, including portfolio optimization, option pricing, and risk management, and cyber-attacks. What kind of opportunities and risks are there? How long do you think it will take for quantum computing to evolve to the point where these issues are relevant?

B. Python exercises

- 6.1 Housing prices
- 6.2 Fraud detection